

Flight Control Circuit Organization in Drosophila BANC Connectome

Major Findings

A. Circuit Architecture

5 Functional Modules Discovered

2,080 Premotor Interneurons Organized into Power and Steering Networks

POWER MODULE

Cluster 4

181 interneurons

STEERING MODULES

Clusters 1, 2, 3, 5

1074 interneurons

Individual Clusters:

- C1: 224 INs → b2, b1, iii4
- C2: 181 INs → DLM1, DLM5, DVM1A
- C3: 139 INs → ps1, iii4, DLM1
- C4: 299 INs → iii4, hg4, hg1
- C5: 412 INs → i2, i1, iii1

Power: Strong connections | Steering: Distributed control

B. Network Modularity

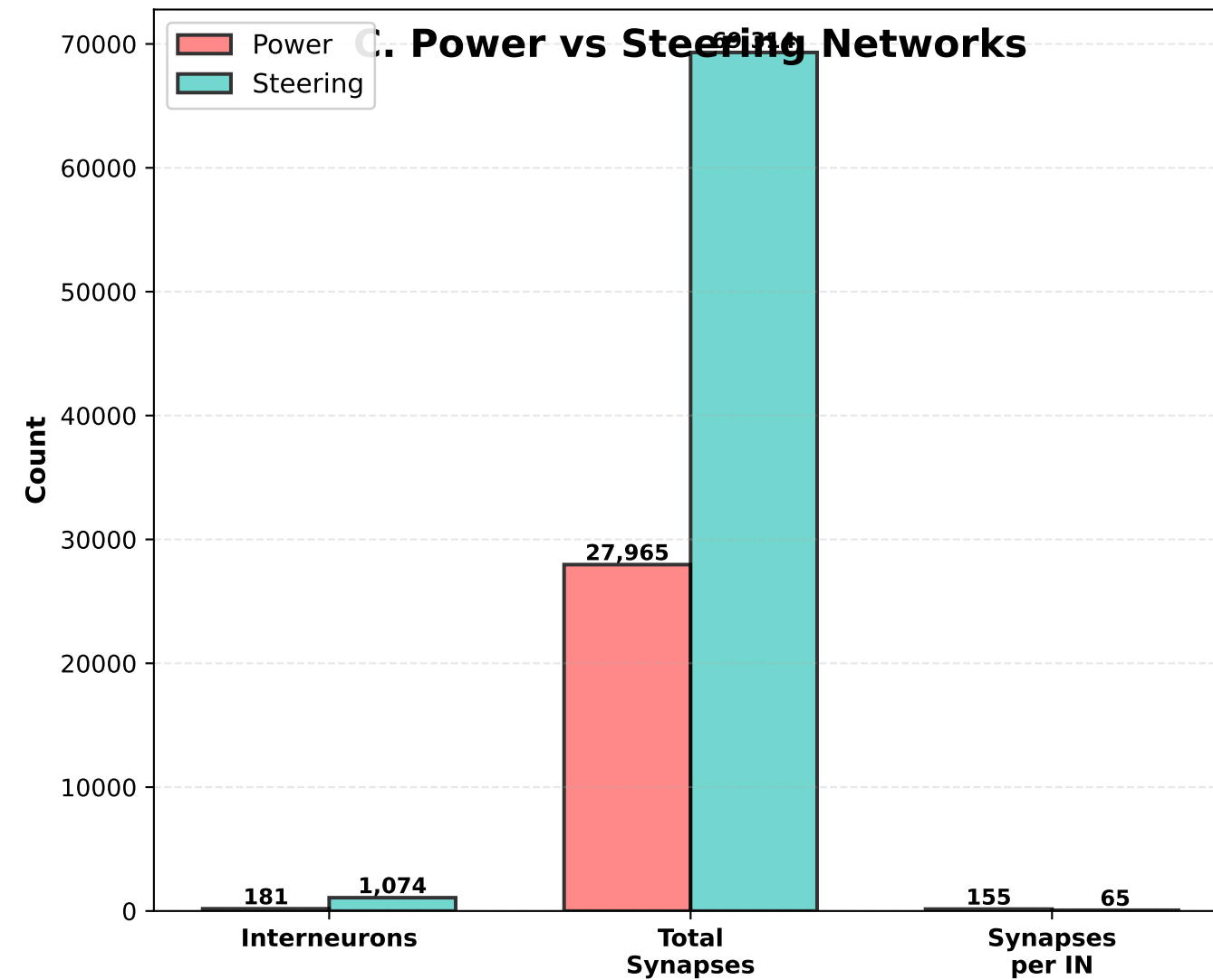
0.510

Modularity Score

MODERATE Functional Separation

Cluster Connectivity Strength:

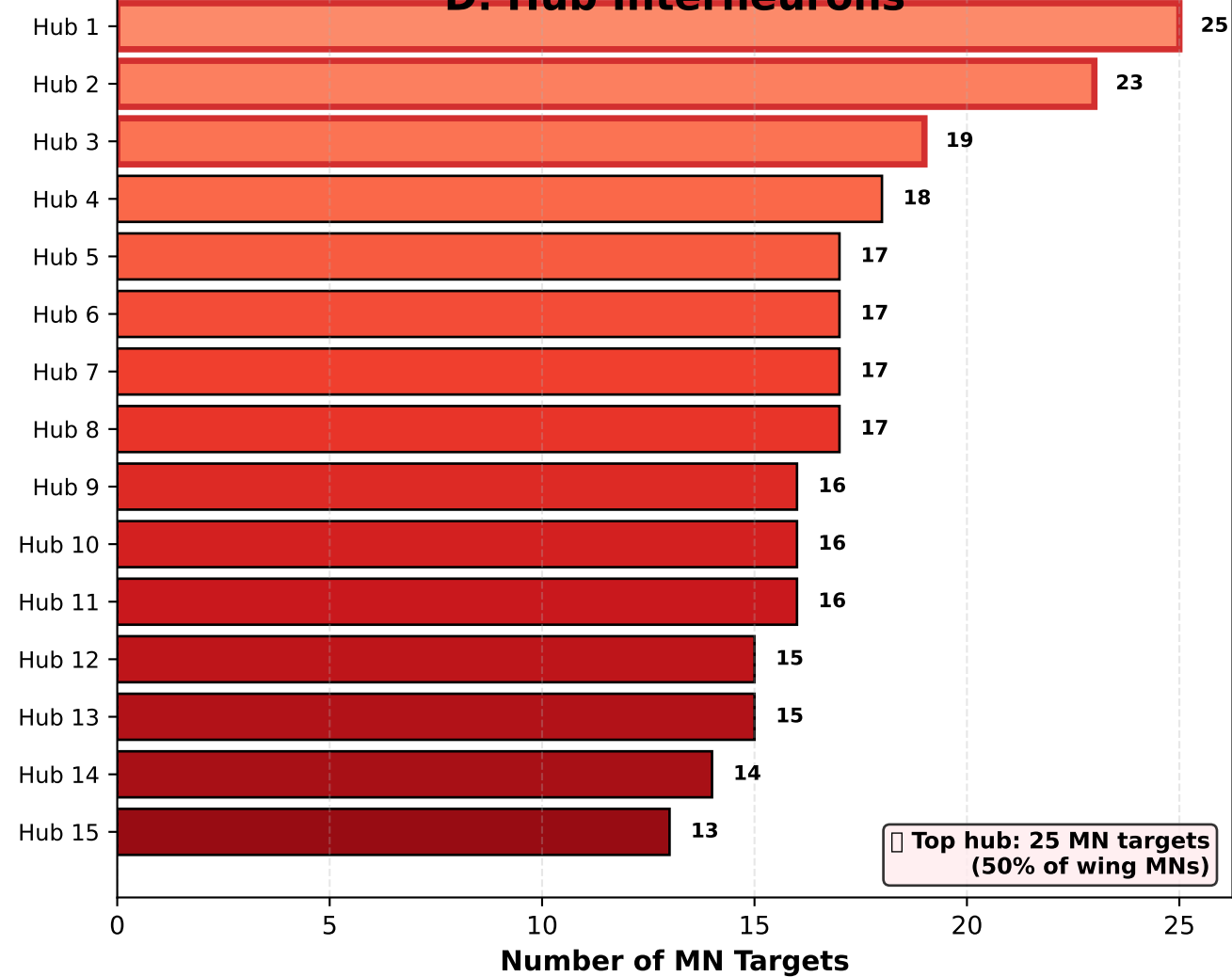
C. Power vs Steering Networks



Steering has 2.5× more total synapses

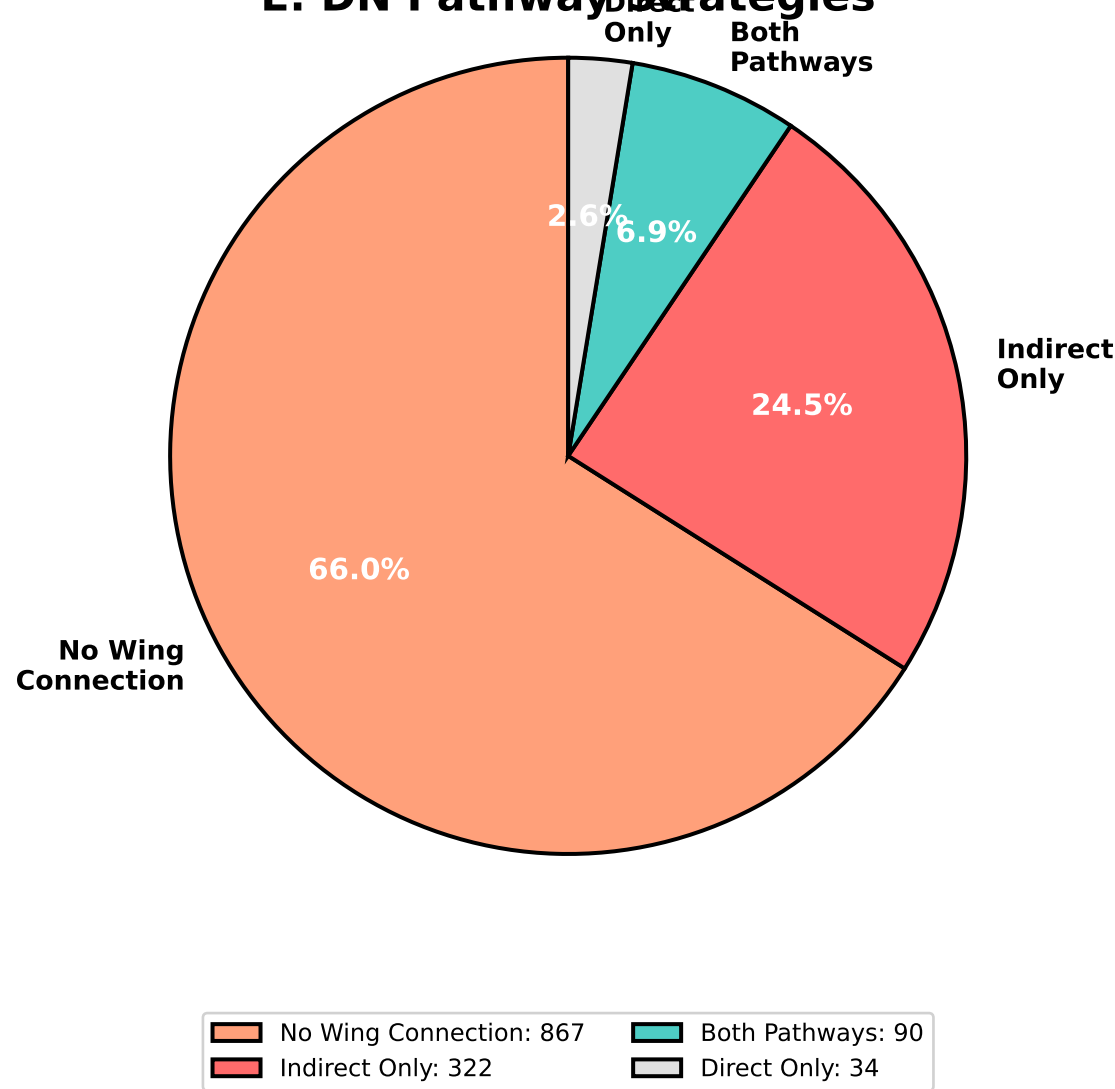
Top 15 of 1,255 Hub Neurons

D. Hub Interneurons



Top hub: 25 MN targets (50% of wing MNs)

E. DN Pathway Strategies



F. Key Statistics

CIRCUIT SUMMARY

Premotor Network:

- Total INs: 1,255
- Connections: 4,370
- Synapses: 97,279

Functional Modules:

- Total: 5
- Power: 1 cluster (370 INs)
- Steering: 4 clusters (1,710 INs)

Hub Neurons:

- Total identified: 1,255
- High centrality: 430
- Max targets: 25

Integration:

- Integrative INs: 58
- Percentage: 4.6%

Descending Neurons:

- Analyzed: 1313
- Both pathways: 90
- Direct only: 34
- Indirect only: 322